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PALEOMAGNETISM OF THE HOG HEAVEN VOLCANIC FIELD, MONTANA: DIFFICULTIES IN AVERAGING PALEOSECULAR VARIATION IN VOLCANIC FIELDS

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Abstract. Twelve virtual geomagnetic poles from 25 paleomagnetic sites in the Oligocene Hog Heaven volcanic field of northwestern Montana yield a mean pole at 74.4°N, 71.5°E (alpha 95 = 5.9°). This pole indicates clockwise rotation of 10.5° ± 7.4° and flattening of 13.2° ± 4.6° at the 95% confidence level. Assuming the axial geocentric dipole hypothesis criteria have been satisfied, at 95% confidence rotation and flattening are statistically different from zero. These results indicate clockwise rotation of the Hog Heaven volcanic field as expected for the Sanpoil block farther west. However under these assumptions this farsided pole requires northward transport of nearly 1400 km, about 100 times too much given the geologic constraints. Thus, although radiometric dates indicate 5 million years of volcanism, several igneous rock types are present, there are interbedded sediments and erosional intervals during volcanism, and both stable states of the geomagnetic field are recorded, this pole is not a reliable estimate of the Oligocene rotation axis. Perhaps the volcanic events were not distributed independently with respect to paleosecular variation, or a long-term nondipole field component persisted during much of the evolution of the Hog Heaven volcanic field. These results and previous studies indicate paleomagnetic data from one volcanic field may not be sufficient to estimate tilt, translation, or rotation of a tectonic block.

Introduction

The Oligocene Hog Heaven volcanic field (Figure 1) of northwestern Montana comprises approximately 50 km^2 of welded and unwelded ash-flow tuffs, related silicic extrusive and intrusive rocks, and volcaniclastic sediments. The volcanic complex, on the crest of the late Cretaceous Purcell anticlinorium, is at the northernmost extreme of Cenozoic Basin and Range crustal extension. The remnants of the volcanic field rest unconformably on Precambrian Y Belt Supergroup argillites and quartzites, these Precambrian rocks were translated more than 100 km to the east during the emplacement of the Purcell anticlinorium. Four K-Ar dates [Marvin et al., 1984] indicate igneous activity from about 36 mybp to 31 mybp.

In northeastern Washington, 400 km west of the Hog Heaven volcanic field, lie the Eocene Sanpoil volcanic rocks. West of the Sanpoil volcanic field, clockwise rotations of Tertiary rocks are common [e.g., Wells and Heller, 1988]. Fox and Beck [1985], expecting that northeastern Washington was east of the region affected by Tertiary rotations, conducted a paleomagnetic study of the Sanpoil volcanic rocks. Their discovery that the Sanpoil volcanic field is rotated 25° clockwise lead them to propose that the rotated Sanpoil block (Figure 1) extends east, beyond the Hog Heaven volcanic field, to the Rocky Mountain front.

The eastern edge of the Sanpoil block most likely is associated with the Tertiary extensional basins of northwestern Montana. The Rocky Mountain Trench, Flathead Valley, Swan Valley, Kishenehn Basin, and similar extensional basins of western Montana have been active since at least the Eocene [Constenius, 1987]. The listric normal faults bounding the basins are active and structurally adequate to accommodate the rotation of the Sanpoil volcanic field. The Hog Heaven volcanic field lies between these basins and the Sanpoil volcanic field.

The Hog Heaven volcanic field is well positioned to determine if the post 40 mybp rotation of the Sanpoil block also affects the Purcell anticlinorium. These rocks appear to be ideal for such a study as: 1) they span 5 million years, 2) there are interbedded sediments, 3) reverse and normal polarities are recorded by the rocks, 4) numerous lithologies are present, and 5) the rocks are relatively flat lying. However, 25 paleomagnetic sites yielded only 12 reliable virtual geomagnetic poles (VGP) from the Hog Heaven volcanic field. Despite the favorable geologic relations, the 12 VGP do not yield a reasonable estimate of the Oligocene rotation axis.

Field and Lab Procedures and Paleomagnetic Results

At each of 25 outcrops, roadcuts, or mine shafts in the Hog Heaven volcanic field, about eight independently oriented 2.5 cm cores were collected using a portable drill, a magnetic compass, and where possible a sun compass. Local deflections of the magnetic compass caused by the outcrop were never more than a few degrees, none of the sites had results suggestive of extensive remagnetization by lightning strikes. There is no large scale magnetic anomaly associated with the Hog Heaven volcanic field or the associated silver mineralization.

The rock types sampled include lava flows, small dacitic intrusions, welded ash flows, and dikes. The Hog Heaven rocks spent the past 30 million years weathering near the earth's surface, they are not particularly well exposed. Given the poor exposure and complex volcanic stratigraphy of the area, efforts to avoid placing two sites in the same cooling unit may not have been 100% successful. The Hog Heaven volcanic rocks are flat lying, only 4 sites have measurable non-zero dip. Perhaps, because of poor exposure, some tilting is not discernible. However, 9 sites give direct (e.g., interbedded sediments, columnar jointing, flattened pumice in ash-flow tuffs) evidence for no dip. Many more, by nearby indicators, indicate maximum dips within a few degrees of horizontal. There is no regionally

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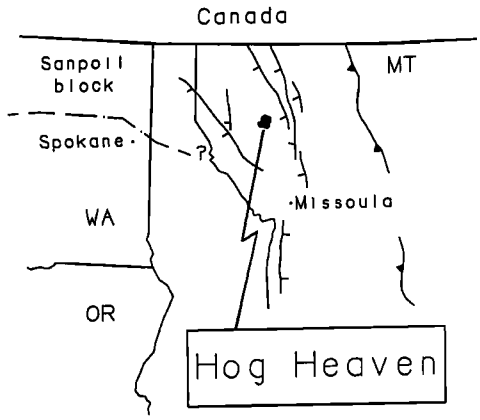


Fig. 1. Location of the Hog Heaven volcanic field and generalized geology of western Montana to eastern Washington.

consistent dip in the Hog Heaven volcanic field. Although allochthonous, the underlying Precambrian Belt sedimentary rocks rest on a flat in the subsurface thrust and are nearly horizontal [Sheriff, 1985], they appear unaffected by the listric normal faults to the east. Regardless, long wavelength gentle dips may have gone unnoticed but the sites are distributed over as large an area as possible to average such mistakes to near zero. The mean directions presented in Table 1 are most likely reliable.

Progressive alternating field (af) and thermal demagnetization techniques were employed to evaluate the natural remanent magnetization (NRM) of the samples. These procedures were performed in two paleomagnetic laboratories. At Western Washington University (WWU), af demagnetization was performed with a Schonstedt GSD-5 tumbling demagnetizer, magnetization was measured with a Schonstedt SSM-1A spinner magnetometer. At the University of Montana (UM) experiments were completed with a Molspin tumbling af demagnetizer and a Schonstedt SSM-2A spinner magnetometer. All thermal demagnetization was completed at UM using a homemade magnetically shielded oven and cooling system with magnetic induction in the sample region less than 6 nT.

Many samples were demagnetized in low inductions at WWU and subjected to additional progressive demagnetization at UM. Most samples showed no change in remanent direction after transport between the labs. Some samples, particularly those demagnetized with inductions greater than about 40 or 50 mT, acquired a significant viscous remanent magnetization (VRM) while between lab sessions. Usually the coercivity of this VRM was similar to the previous peak induction experienced by the sample. Rarely the coercivity of these VRMs was greater than the previous peak induction and greater than the peak demagnetizing capacity (100 mT) of the Molspin af demagnetizer. If these latter samples had not been previously (WWU) demagnetized to a few percent of their NRM they were excluded from further statistical analysis.

Each sample was progressively demagnetized until less than a few percent of its NRM remained or the noise level of the magnetometer was exceeded. For af demagnetization, steps ranged from 5 to 10 mT, up to a peak induction of 100 mT and removed less than 10% of the NRM per step. Some samples carried a significant component of NRM beyond demagnetization at 100 mT, these were then treated to thermal demagnetization. In addition, one or two samples from many sites were thermally demagnetized to determine if thermal demagnetization yielded the same characteristic remanent direction as af demagnetization, such was almost always the case. Thermal demagnetization, in steps ranging from 20°C to 100°C, was carried out until at least 95% of the NRM was unblocked. The peak temperature applied to a specimen depended on its lithology, but always exceeded 625°C to test for the presence of remanent magnetization residing in titanohematites. Some rocks did not physically survive thermal demagnetization above a few hundred degrees.

Components of magnetization isolated during demagnetization were identified using standard orthogonal demagnetization plots and their directions determined using principal component analysis. Directions, determined from

Table 1. Site-mean paleomagnetic data from the Hog Heaven volcanic field.

Site	N/N_c	Dec	Inc	A_{95}	k	Plat	Plong
2	10/10	359.7	50.5	3.2	224.38	73.4	66.5
3	8/13	162.5	-62.0	7.6	54.67	-77.0	321.3
10	7/8	1.5	39.8	5.7	113.58	64.7	62.2
12	6/6	354.2	48.3	8.1	69.68	70.9	80.9
13	6/7	356.5	66.1	6.6	103.73	87.6	172.5
16	7/8	161.6	-59.2	7.2	71.81	-74.6	311.1
20	7/7	1.6	38.1	7.1	73.87	63.5	62.1
21	5/6	358.6	42.4	4.5	294.21	66.6	68.6
22	6/7	177.9	-50.6	9.4	52.14	-73.4	251.8
23	7/7	8.3	45.9	2.7	495.18	68.4	44.9
24	7/8	4.4	57.9	1.9	1006.15	80.1	45.0
25	8/8	4.6	46.4	4.3	169.67	69.5	53.6

Site is the site number, N/N_c is the number of specimens used/collected, Dec and Inc are the site mean declination and inclination, A_{95} is the semiangle of 95% confidence cone in degrees, Plat and Plong are the latitude (positive north) and longitude (positive east) of the virtual geomagnetic pole. The center of Hog Heaven is at 48°N, 245.5°E.

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three or more demagnetization steps, always had maximum angular deviations less than 12° and usually less than 5°. Site mean directions were calculated by the method of Fisher (1953).

Twelve of the original 25 sites yield reliable site mean directions, 13 sites were excluded from the final statistical analysis for the following reasons. Six rejected sites were in highly altered rocks associated with the silver-based mineralization that accompanied late stage volcanism at Hog Heaven. Samples from these six sites did not yield reliable characteristic remanent magnetizations regardless of thermal or af demagnetization experiments, apparently the alteration removed any original titanomagnetites. Three sites were in volcaniclastic rocks that did not record consistent within-site directions. One rejected site, in a blocky dacite flow, apparently acquired magnetization while individual blocks were in motion relative to each other. Three sites yielded virtual geomagnetic poles more than 50° from the mean of the other sites, most likely these sites acquired magnetization during a reversal or anomalous behavior of the geomagnetic field.

The 4 sites with measurable dip were among the 13 rejected. The average direction of the remaining 12 sites (Table 1) has declination = 358.5°, inclination = 50.8° (the 95% circle of confidence, alpha 95 = 5.5°, Fisher's [1953] estimate of the precision parameter, k = 63.5).

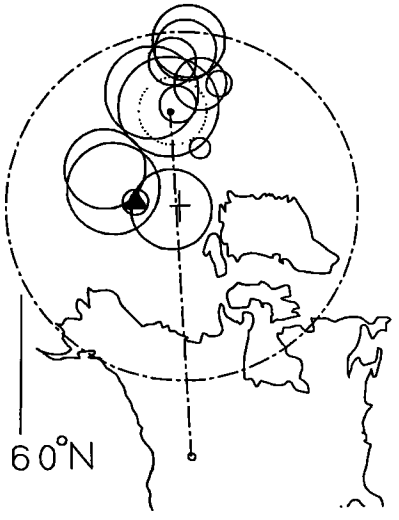


Fig. 2. Stereonet showing the circles of 95% confidence for the 12 VGP at their centers. The mean pole, connected to Hog Heaven with the dashed line, is shown by the dotted 95% confidence circle. The plus sign shows the spin axis, the 95% confidence circle with the inscribed triangle is the reference direction (Diehl et al., 1988).

Discussion

Diehl et al. [1988] provide a refined paleomagnetic reference pole, for North America between 20 and 40 mybp, at 81.5°N, 147.3°E, alpha 95 = 2.4°. Their pole includes 239 virtual geomagnetic poles [VGP] from 16 different studies. Although Diehl et al. [1988] include two unit means from extremely extended terrain in western Nevada [see Geissman et al., 1982, and Proffett, 1977], removal of theses two VGP would not significantly change the result. Their result is most likely an accurate estimate of the mid-Tertiary pole position. The mean of the 12 reliable VGP (Figure 2) from the Hog Heaven volcanic field is at 74.4°N, 71.5°E, alpha 95 = 5.9°. The Hog Heaven pole is not coincident with the reference pole at the 95% confidence level. The Hog Heaven pole is east of the reference pole and, relative to the sampling location, on the far side of both the reference pole and current spin axis (Figure 2).

The expected mid-Tertiary declination and inclination, calculated relative to the reference pole, at the Hog Heaven site is D = 348°, I = 64°. The measured direction is D = 358.5°, I = 50.8°. Calculating the apparent rotation and flattening (with 95% confidence limits) by Beck's [1980] method with Demarest's [1983] correction indicates rotation of 10.5° ± 7.4° and flattening of 13.2° ± 4.6°. Assuming the axial geocentric dipole hypothesis criteria have been satisfied, at 95% confidence rotation and flattening are statistically different from zero. These results indicate clockwise rotation as expected for the Sanpoil block. However, under these assumptions, they require northward transport of nearly 1400 km, this much transport is geologically unreasonable. The Hog Heaven volcanic rocks pierce the Precambrian Belt Supergroup and stitch the deep thrust which carries the Belt rocks and the Paleozoic and Mesozoic section to the east. Motion on the thrusts ended in the Early Eocene, subsequent extension on listric normal faults allows for rotations about nearby vertical axes. These regional relationships and their palinspastic restoration restrict the possibility of northward translation to a few percent of that required by the most probable flattening. There is no regional dip sufficient to account for the discrepancy. More likely, the Hog Heaven volcanic field did not sample the secular variation of the Oligocene geomagnetic field sufficiently to yield an accurate estimate of the rotation axis.

We expect volcanic fields like Hog Heaven to adequately average paleosecular variation to a constant value. The radiometric dates imply a duration of five million years, several igneous rock types are present, there are interbedded sediments, some flows fill ancient valleys suggesting erosional intervals, both stable states of the magnetic field are recorded in the rocks. However, the circular standard deviation of the 12 VGP from 25 sites is only 10.9° [expect ≈ 17°, McFadden et al., 1988].

There are at least two components in any explanation for the failure of the Hog Heaven volcanic field to adequately average paleosecular variation despite the geologic observations. First, perhaps the volcanic events were not distributed independently with respect to paleosecular variation, rather the volcanic field may have developed during a small number of active periods. Second, long-term nondipole components of the geomagnetic field [e.g., Schneider and Kent, 1988] may persist for periods approaching a significant portion of the duration of igneous activity at Hog Heaven. At Hog Heaven there are too few radiometric dates of adequate precision, insufficient outcrop, and too limited stratigraphic continuity to separate contributions from these two mechanisms. It is doubtful one could reconstruct a reliable history of igneous activity even with more dates of greater precision [see Zehner, 1986]. Adequate rocks do not exist to magnetostratigraphically bound the duration of volcanism.

The above problems are not unique, many studies assume their results average secular variation to a reliable constant value when it may not be true. Some apparently episodic volcanic fields fail to randomly sample the polarity record during their duration as measured by radiometric or

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stratigraphic methods. Two modern detailed and comprehensive studies of volcanic rocks indicate how difficult it can be to determine an accurate paleomagnetic pole. First, Diehl et al. [1988] have 239 mid-Tertiary VGP from 16 different studies, yet the reverse and normal polarity subsets yield paleomagnetic poles that are different at the 95% confidence level. Second, Calderone et al. [1988] estimate that they obtained only 65 VGP from 190 middle Miocene lava flows exposed in the Mojave-Sonora desert of Arizona and California. Limited subsets of either of these two studies would not always yield an accurate pole [cf., Calderone and Butler, 1984]. These considerations, the Hog Heaven results, and recent studies of paleosecular variation prove it may be difficult to make accurate estimates of the ancient spin axis from reasonably extensive volcanic fields.

The paleomagnetic data obtained from the Hog Heaven volcanic field are not adequate to define the eastern limits of the Sanpoil block. Although permissive of rotation far too much northward transport is indicated. The best interpretation is that the Hog Heaven results do not provide an accurate estimate of the Oligocene rotation axis. The lack of better resolution is due to the geologic record in the Hog Heaven volcanic field. Nearly 25% of the 25 available paleomagnetic sites were adversely affected by late-stage silver-based alteration and mineralization, only 12 sites yield reliable VGP.

To estimate tilt, translation, or rotation of an individual terrane or tectonic block by analysis of paleomagnetic poles or mean directions requires a sufficiently long and sufficiently continuous record of the geomagnetic field. Given the Hog Heaven data and the above examples, one should use more than one volcanic field or assemblage per terrane or tectonic block. This tends to restrict the use of paleomagnetism concerning local palinspastic restorations based on analysis of poles or mean directions. Thus, models become more regional than the often desired local palinspastic restoration but they will be a more accurate contribution to our understanding of processes. Studies of individual cooling units [e.g., Sheriff 1984; Wells and Hillhouse, 1989] which allow measurement of tilt or rotation at many scales are best for local palinspastic reconstructions.

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